# Impulse Response and LTI Systems - Part II

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# Key Concepts

1) The impulse response $h[n]$ is the output of a system when the input is an impulse $\delta[n]$.

2) If the system is time invariant, then it responds the same whenever the input is applied. Thus, applying a delayed input $\delta[n-l]$ produces a delayed impulse response $h[n-l]$.

3) If the system is also linear, then the output in response to a weighted sum of input signals is the weighted sum of the individual output signals. So an input $x[n] = a\delta[n] + b\delta[n-l]$ generates an output $y[n] = ah[n] + bh[n-l]$.

4) We write an arbitrary input as a weighted sum of time-shifted impulses

$$x[n] = \sum_{k=-\infty}^{\infty} a_k \delta[n-k]$$

where $a_k = x[k]$, the signal amplitude at time $k$. For LTI systems the output is the weighted sum of time-shifted impulse responses. Since $a_k \delta[n-k]$ produces an output $a_k h[n-k]$, we have

$$\begin{aligned} y[n] &= \textstyle\sum_{k=-\infty}^{\infty} a_k h[n-k] \\ &= \textstyle\sum_{k=-\infty}^{\infty} x[k] h[n-k] \\ &= x[n] * h[n] \end{aligned}$$

This expression for the output is called the convolution sum.

5) Properties of convolution:

a) Commutative: $x[n] * h[n] = h[n] * x[n]$.

b) Cascade systems (associative): $(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$.

c) Parallel systems (distributive): $x[n] * h_1[n] + x[n] * h_2[n] = x[n] * (h_1[n] + h_2[n])$.

6) A system is bounded-input bounded-output (BIBO) stable if and only if the output is bounded for any bounded input.

7) An LTI system is BIBO stable if and only if the impulse response

is absolutely summable

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

8) The output $y(t)$ of an LTI continuous-time system with input $x(t)$ and impulse response $h(t)$ is given by the convolution integral

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

9) Continuous-time LTI systems have properties analogous to discrete-time LTI systems

a) Commutative: $x(t) * h(t) = h(t) * x(t)$.

b) Cascade systems (associative): $(x(t) * h_1(t)) * h_2(t) = x(t) * (h_1(t) * h_2(t))$.

c) Parallel systems (distributive): $x(t) * h_1(t) + x(t) * h_2(t) = x(t) * (h_1(t) + h_2(t))$.

d) A system is BIBO stable if and only if the impulse response is absolutely integrable

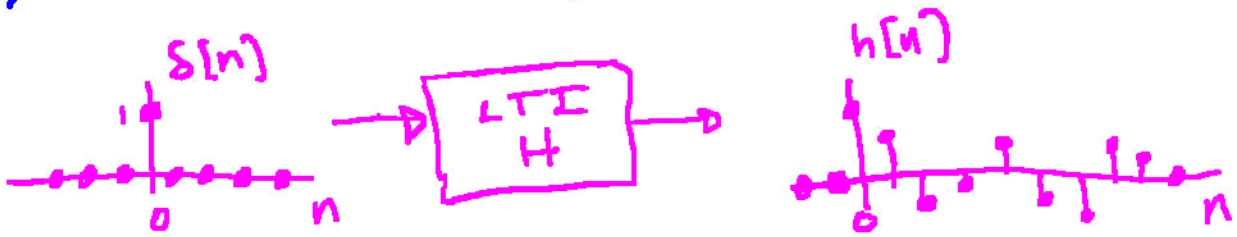
$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

# Video Screenshots

# Relate output of LTI system to input



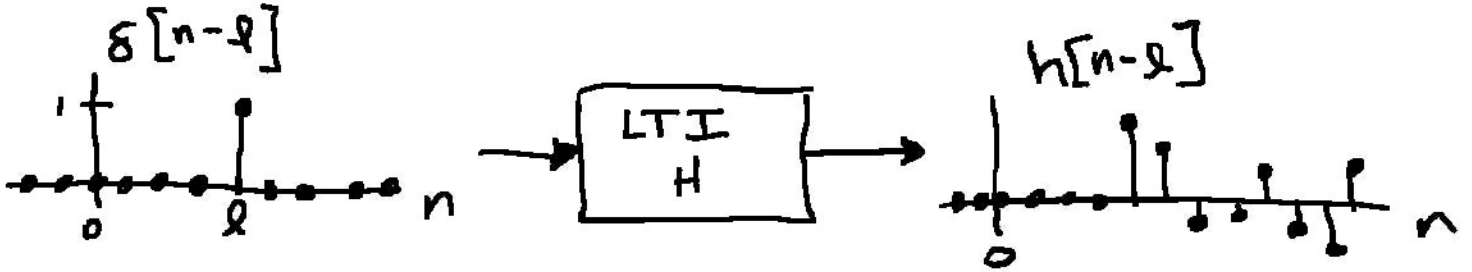
Impulse response

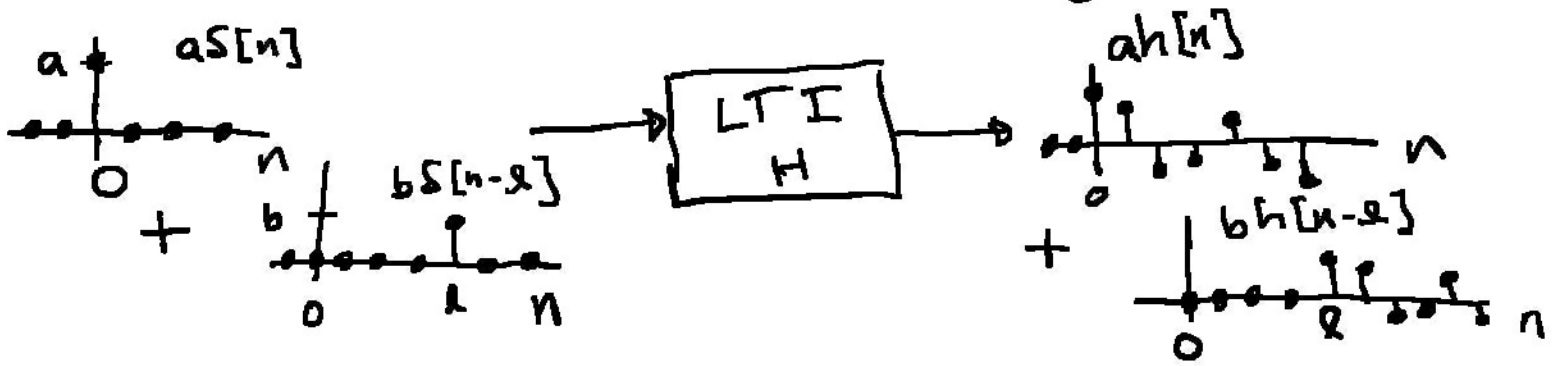



Time invariance

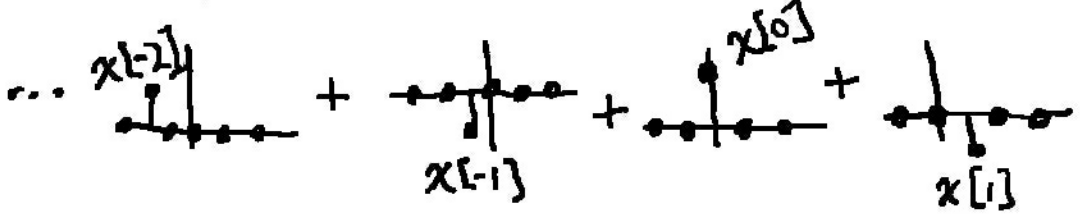
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Linearity




$$x[n] = \sum_k x[k]\,\delta[n-k]$$







$$y[n] = \ldots x[-2]\,h[n+2] + x[-1]\,h[n+1] + x[0]\,h[n] + x[1]\,h[n-1] + \ldots$$

$$y[n] = \sum_k x[k]\,h[n-k]$$

## Convolution

$$y[n] = \sum_{l=-\infty}^{\infty} x[l]\,h[n-l] = x[n] * h[n]$$

## Commutative:

let $k = n - l$

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\,x[n-k] = h[n] * x[n]$$

## Cascade of systems:




$$y[n] = \sum_{l} w[l]\,h_2[n-l] \;;\; w[l] = \sum_{k} x[k]\,h_1[l-k]$$

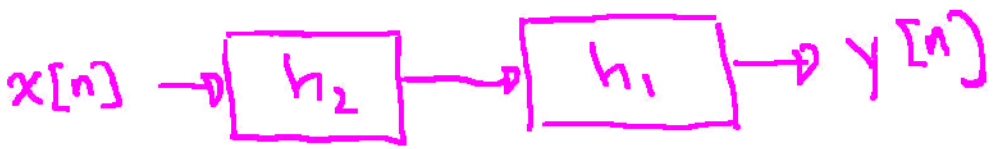
$$y[n] = \sum_{l}\sum_{k} x[k]\;h_1[l-k]\,h_2[n-l]$$

let $p = l - k$

$$= \sum_{k} x[k] \underbrace{\sum_{p} h_1[p]\,h_2[(n-k)-p]}$$

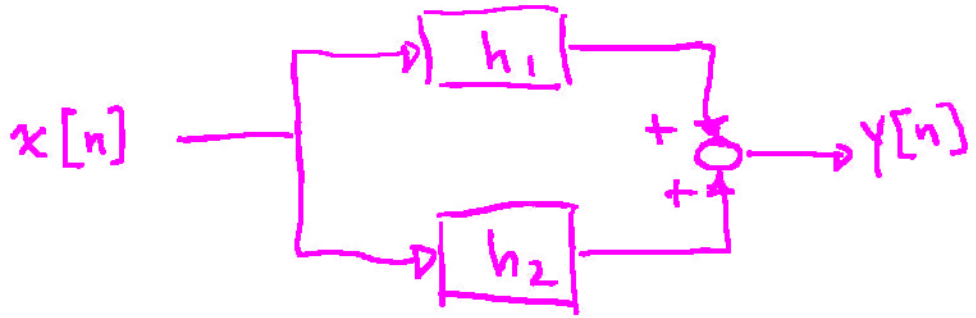
$h[n-k] \Rightarrow h[n] = h_1[n] * h_2[n]$

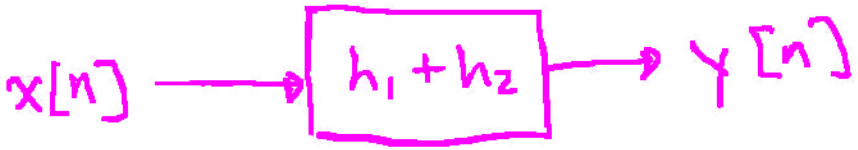


# Parallel systems






$$y[n] = \sum_k x[k]\, h_1[n-k] + \sum_k x[k]\, h_2[n-k]$$

$$y[n] = \sum_k x[k] \underbrace{\left(h_1[n-k] + h_2[n-k]\right)}_{h[n-k]}$$

$$= x[n] * h[n]$$

where $h[n] = h_1[n] + h_2[n]$

Stability: if the input is bounded, a stable system will have a bounded output (BIBO) 4

$$y[n] = \sum_k h[k]\, x[n-k]$$

Suppose $|x[n]| \le C$, (bounded input) consider $y[0] = \sum_k h[k]\, x[-k]$

What $x[n]$ will make $y[0]$ as big as possible?

1) Need all $h[k]x[-k] > 0$

2) max value of $h[k]x[-k] = |h[k]|\, C$ when $x[-k] = C \operatorname{sign}\{h[k]\}$

$$|y[0]| \le C \sum_k |h[k]|$$ bounded if $\sum_k |h[k]|$ is bounded

BIBO Stable $\iff$ $\sum_{k=-\infty}^{\infty} |h[k]| < \infty$

# Continuous-time LTI Systems

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$ "convolution"

## Commutative

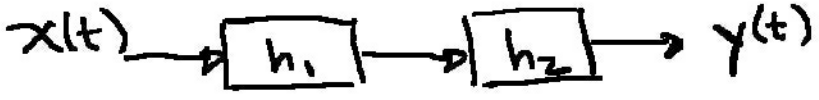
$$y(t) = x(t) * h(t) = h(t) * x(t)$$

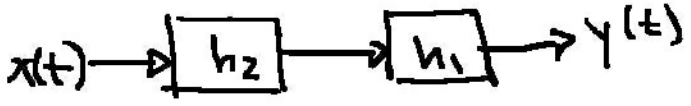



## Cascade

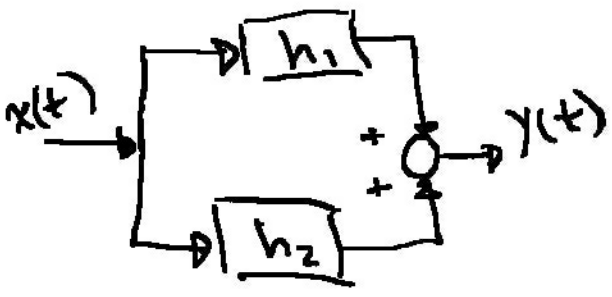
$$y(t) = x(t) * \underbrace{h_1(t) * h_2(t)}_{h(t)} = x(t) * h(t)$$

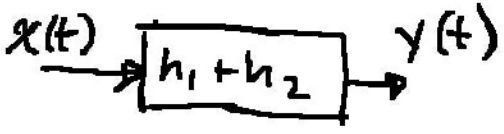
$$= x(t) * h_2(t) * h_1(t)$$




## Parallel




## Stability: Impulse response $h(t)$

BIBO $\int_{-\infty}^{\infty} |h(t)|\, dt < \infty$

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